

History-Augmented Temporal Attention for Smooth Waypoint Handoff in NMPC Policy Distillation



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Abstract

- NMPC** (Nonlinear Model Predictive Control) gives accurate robot path-following commands but can be too expensive for limited onboard hardware.
- We distill the NMPC first receding-horizon command into a compact student policy.
- Problem: feedforward students can imitate accuracy yet produce abrupt waypoint handoff behavior.
- Research Question: **what handoff benefit, computational cost arise when a history policy bundle trained from simulated NMPC data is deployed on a physical robot?**
- Proposed bundle: two-waypoint temporal attention, eight-step GRU history summary, and history dropout.
- Temporal attention reported median **translational handoff jump decreases from 0.339 to 0.281 m/s**, approximately 17%.
- However, the temporal bundle adds 0.3865 ms at p95, reaching 1.1451 ms on an RK3328 CPU under a 5 ms budget.
- Best outcome: smoothest handoff, lowest mean CTE RMSE, zero handoff failures, and p95 latency of 1.1451 ms.**

Introduction / Preliminary Tests

- NMPC **computation** requirements can often exceed the budget of limited hardware.
 - We make an optimal neural-network policy to imitate NMPC accuracy + smoothness
 - Top-down analysis: we **verify whether the NMPC teacher command responds to g_2** before the switch. Then, compare Temporal Attention to the four other architecture bundles, and report implementation cost.
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- Waypoint handoff requires reasoning over current waypoint g_1 and next waypoint g_2 before the switch.
 - Teacher audit: rotating g_2 changed the initial NMPC command at many fixed anchors.
 - Useful response rate: 55.1% pre-switch and 61.7% post-switch using a 0.01 m/s floor.

Region	N	Median diff (m/s) [Q1,Q3]	Above Threshold
Pre-switch	236	0.0144 / [0.004, 0.057]	130/236 (55.1%)
Post-switch	264	0.0187 / [0.006, 0.088]	163/264 (61.7%)

Next-waypoint sensitivity test confirms that g_2 affects the teacher command before handoff.

Poster focus

- Compact policy for resource-constrained robots.
- Smooth transition at waypoint switch.
- Physical evaluation, not only offline loss.

Methodology

Student-policy design

- Inputs: current state x_t , active waypoint g_1 , next waypoint g_2 , phase z_t , and eight history vectors.
- Attention: current-state query over two waypoint keys/values.
- Temporal memory: 64-unit GRU summarizes normalized delta-pose history $[\Delta x, \Delta y, \Delta \theta]$.

Implementation cost

- Temporal model: 338,205 parameters; 1.29 MiB file; zero deadline misses in 1,000 retained calls.

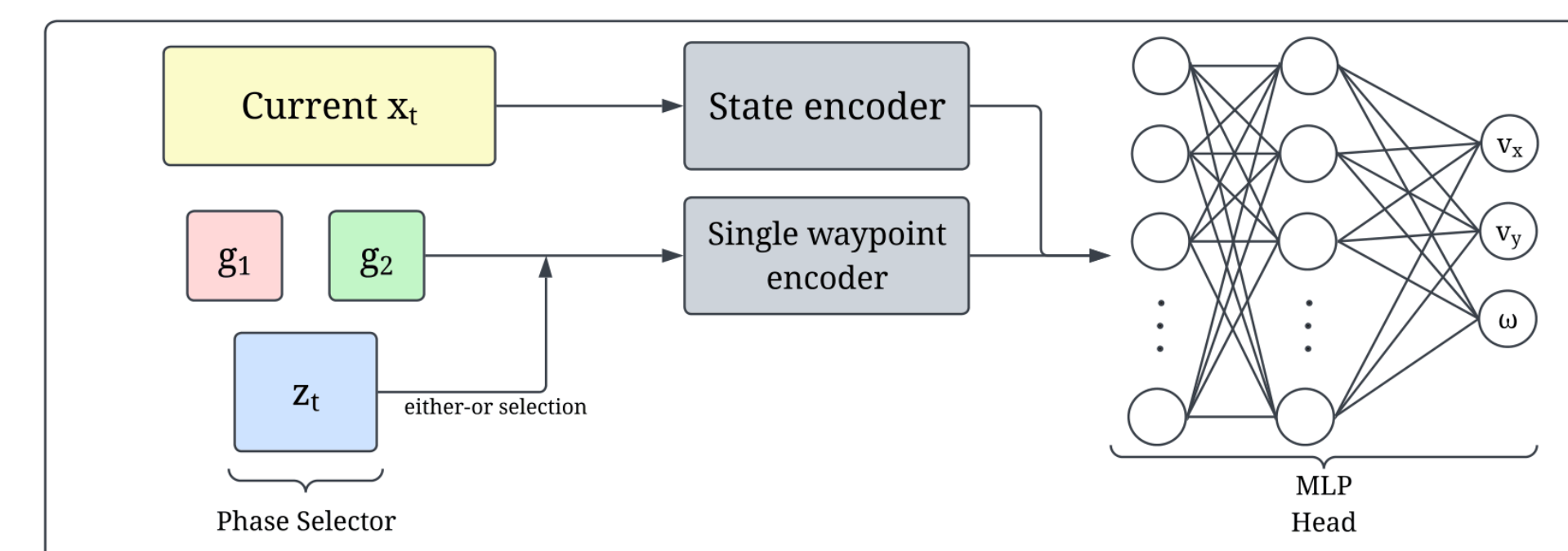
Evaluation protocol

- Baselines: Hard Selector, Joint MLP, Scalar Gate, and Lightweight Attention.
- Training: deterministic NMPC data; 57,000 samples from 3,000 routes.
- Deployment: 4 routes $\times$ 3 seeds $\times$ 5 architectures = 60 physical route-seed cells on RK3328 CPU.
- Metrics: handoff failure, completion, peak jump, CTE RMSE, latency p50/p95, and deadline misses.

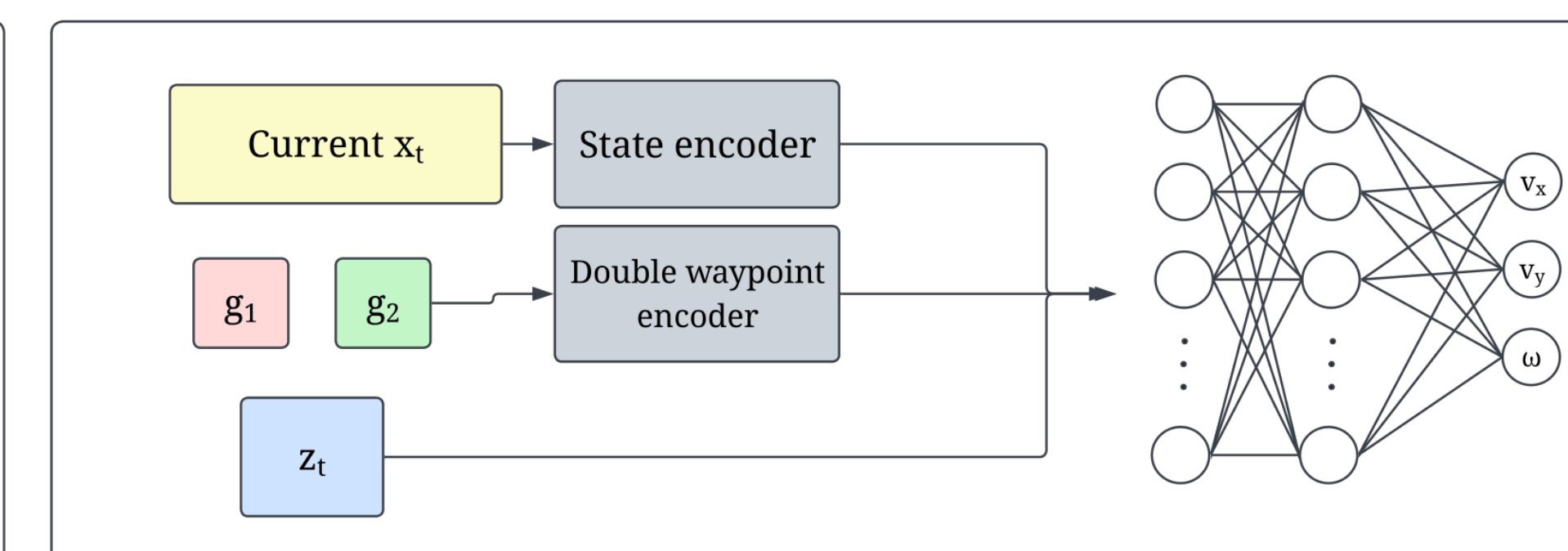
Architecture	Params / file size	Handoff fail	Median jump (m/s)/[min, max]	Completion	Mean CTE RMSE (m)	Latency p50/p95; misses/1000
Hard	138,355 / 540 KiB	1/12	0.377 [0.177, 0.662]	11/12	0.0594	0.2896 / 0.3920; 0/1000
Joint MLP	121,683 / 475 KiB	0/12	0.383 [0.143, 0.610]	12/12	0.0618	0.2325 / 0.3535; 0/1000
Scalar Gate	138,420 / 540 KiB	0/12	0.382 [0.178, 0.647]	11/12	0.0526	0.3870 / 0.4888; 0/1000
Lightweight Attention	269,351 / 1.03 MiB	0/12	0.339 [0.142, 0.609]	12/12	0.0481	0.6277 / 0.7586; 0/1000
Attention w/ History GRU	338,205 / 1.29 MiB	0/12	0.281 [0.152, 0.460]	12/12	0.0468	1.0421 / 1.1451; 0/1000

Full table of results, including failures, file size, smoothness, and mean cross-track RMSE error

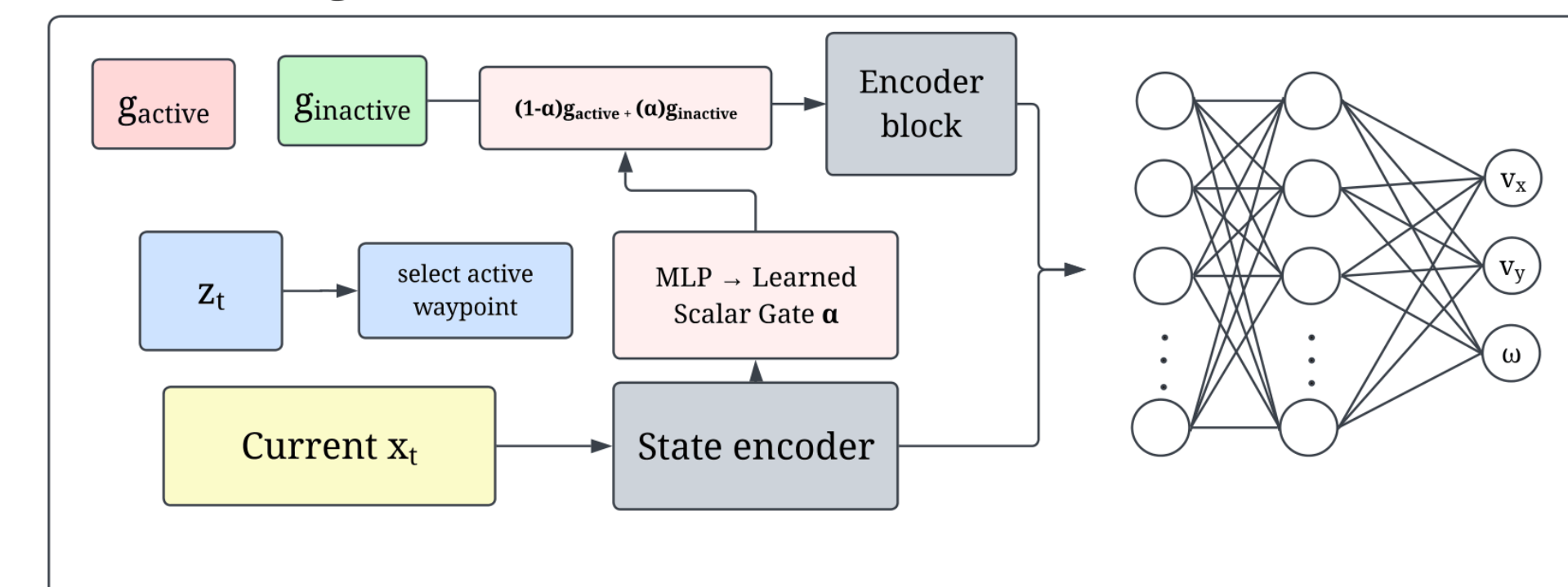
Hard Selector



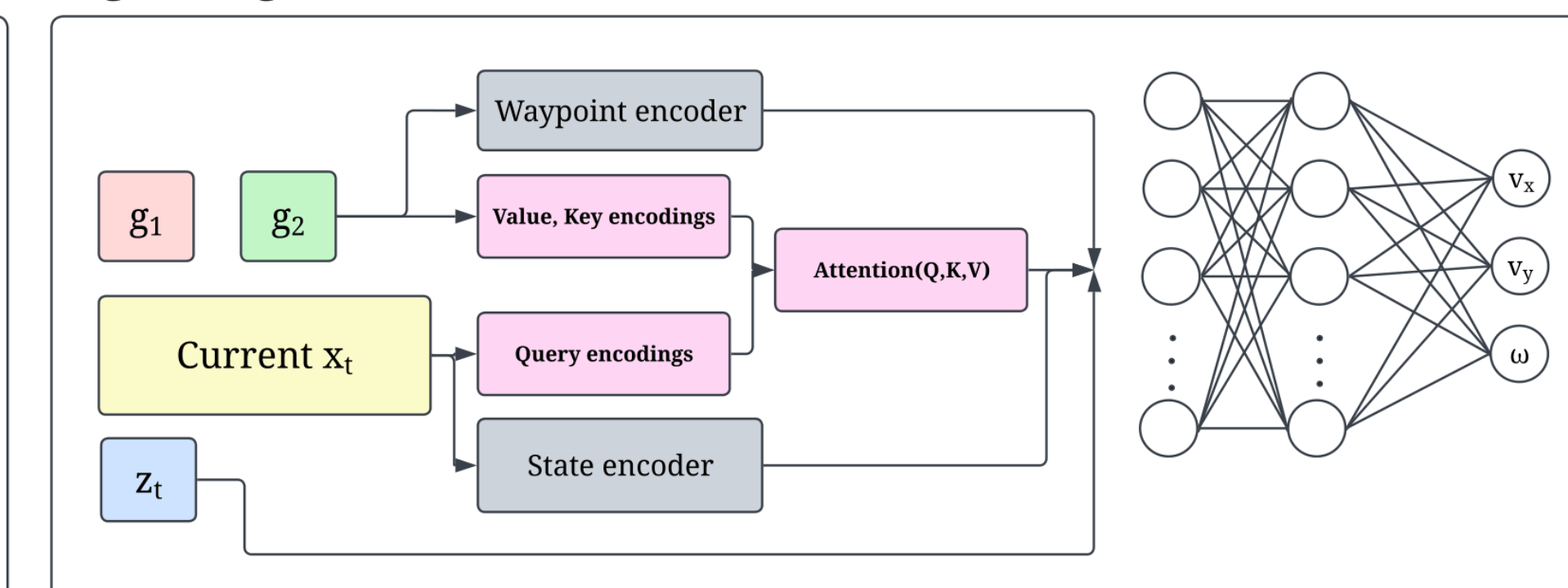
Joint MLP



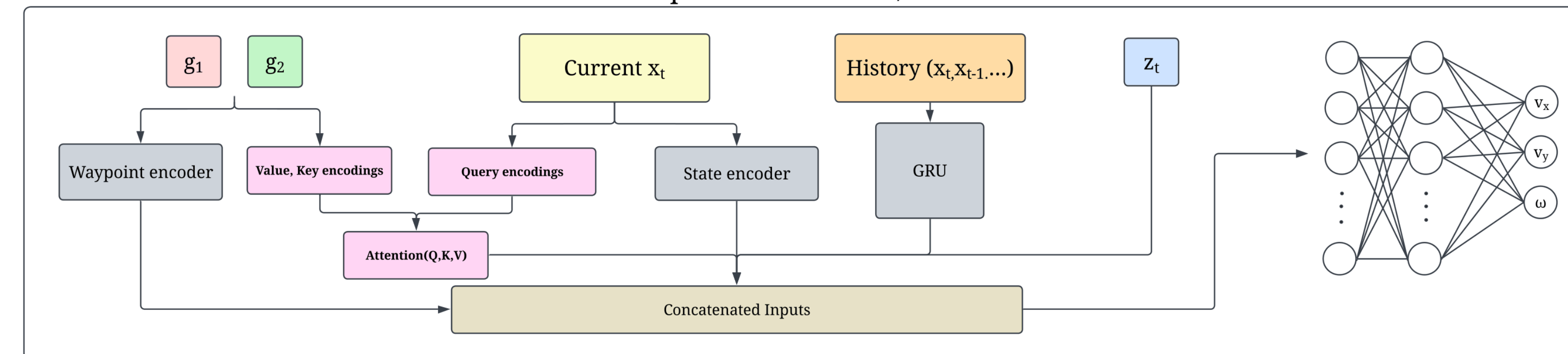
Scalar Gating



Lightweight Attention



Temporal Attention w/ GRU

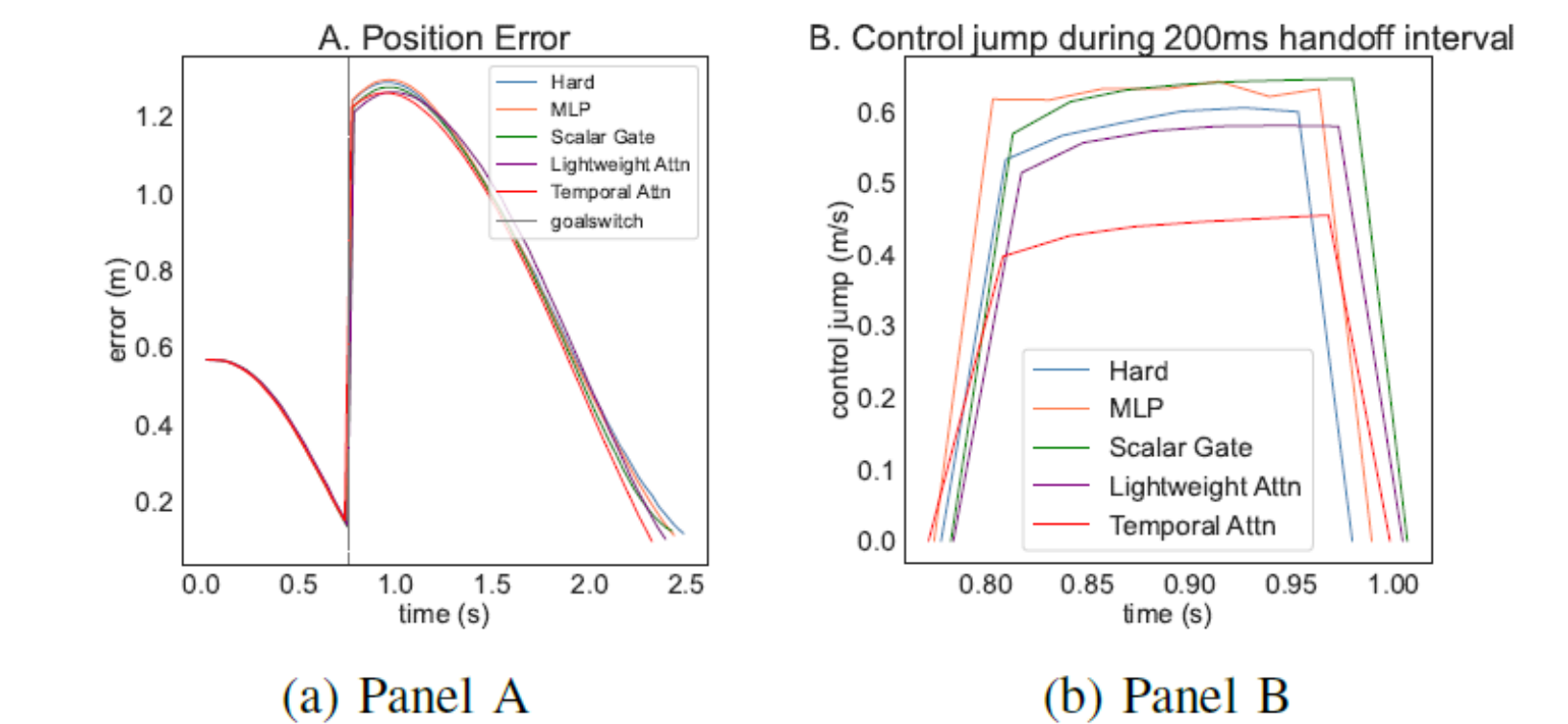


Five candidate NMPC student architectures. Temporal Attention adds GRU history to the two-waypoint attention pathway.

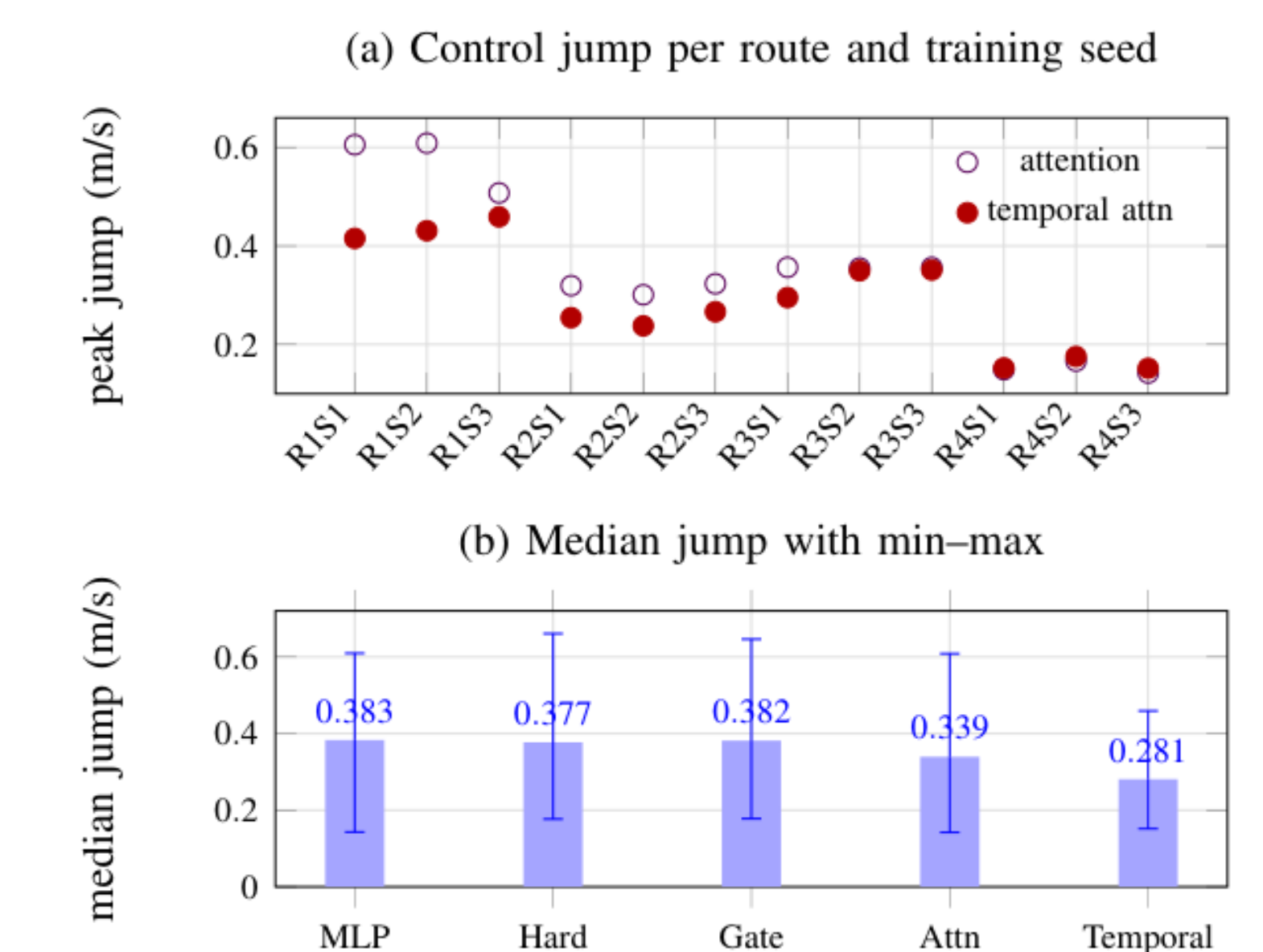
Results

Closed-loop physical results

- Temporal Attention completed all 12 runs with zero handoff failures.
- Median handoff jump: 0.281 m/s – lowest of all five models.
- Lowest mean CTE RMSE: 0.0468 m.
- p95 latency: 1.1451 ms; 0/1000 misses under the 5 ms ceiling.



Representative physical trace. Panel A – position error; Panel B – control jump 200ms after handoff



Control-jump comparison: temporal attention lowers aggregate median jump and reduces jumps on Routes 1-3.

Conclusion

- History helps, since the GRU summary adds temporal context that smooths waypoint handoff.
- Attention helps, since the policy can query both g_1 and g_2 instead of hard-switching between goals.
- Temporal Attention is the strongest tested student policy for smooth two-waypoint NMPC handoff, supports future evaluation of temporal policies for NMPC-sim-to-real transfer.

Limitations + next steps

- We introduce temporal GRU + history dropout to prevent overfitting. However, future research should run ablation studies for evaluation of recurrence vs. history dropout.
- Future work can train with disturbance, sensor-noise, and drift models; and pair with safety or stability filtering.